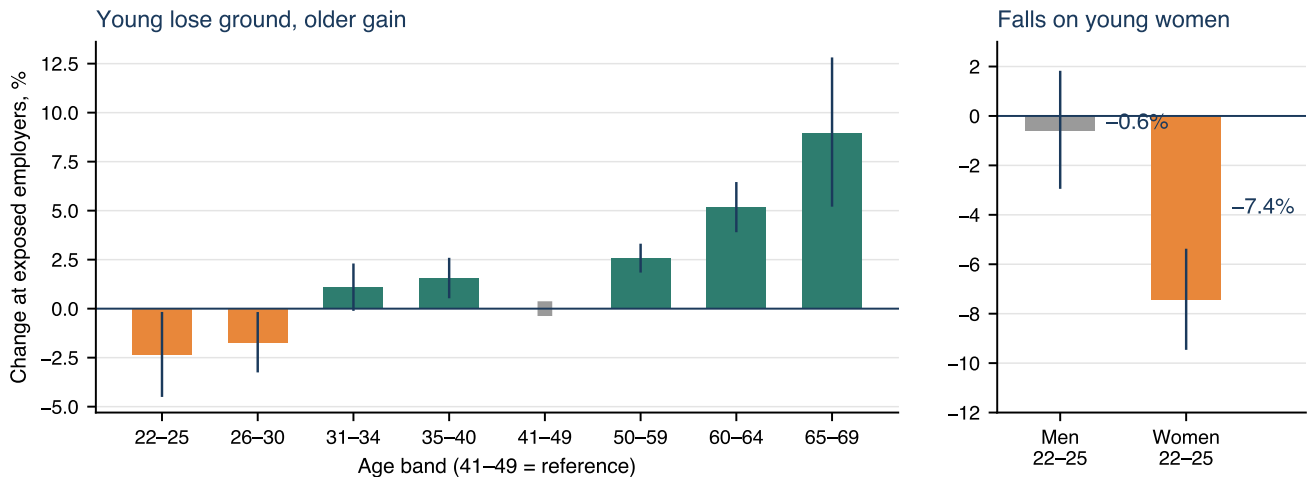


Same Storm, Different Boats: Generative AI and the Age Gradient Within Firms

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Key finding: At Swedish employers most exposed to generative AI, the number of **22–25-year-olds** relative to their older colleagues fell **3.9 per cent** more from 2023 to 2024–25 than at less exposed employers. The shortfall falls on **young women**, and the relative gain **rises with age**.



Change from 2023 to January 2024–June 2025 at the most exposed quarter of employers, compared with less exposed employers. Left: each age band against workers aged 41–49 at the same employers. Right: young women and young men, each against older colleagues of the same sex. Lines: 95 per cent confidence intervals.

What we find

- **Two shocks, one date apart.** Job postings fell after 2022 while share prices rose. The Riksbank's first rate rise came seven months before ChatGPT. Which of the two dates the posting decline in exposed occupations follows depends on how exposure is measured.
- **Employers shift their advertising.** Within the same employer, advertisements for exposed occupations fall about 15 per cent relative to its other advertisements after ChatGPT.
- **The young lose ground inside exposed employers.** The young-to-older employment ratio falls 3.9 per cent more at ages 22–25 and at ages 26–30. The relative gain rises with age, to 8.9 per cent at 65–69 against 41–49.
- **Young women bear the shortfall.** Young women lose 7.4 per cent against their older colleagues; young men do not change. The gap is measured inside the same employers and survives largely within fields of education.
- **Separations rise, hires do not fall.** Relative separations of the young rise, though imprecisely once industries may differ. More than half of employees

under 25 hold fixed-term contracts, so a contract not renewed counts as a separation.

How much weight it bears

- The result describes **who works at exposed employers**, not how many jobs exist in the economy.
- It largely holds against industry-wide movements, firm debt (the credit channel of tightening), remote work and pension-age changes. On an alternative exposure measure the women's result is unchanged and the pooled estimate less precise.
- The ratio also moved in the pandemic year, but then it fell on young men.

Why it matters for policy

- **Watch composition, not only totals.** A reshuffling inside employers is invisible in aggregate employment and posting counts.
- **Entry routes matter.** Fixed-term first jobs are where the adjustment shows; work-integrated learning and vocational routes (YH) that build experience with AI tools deserve attention, for young women in particular.
- **Timely occupational statistics.** Occupation codes arrive about two years late, and on their own they can manufacture a spurious youth decline.

Data and method: Monthly employer declarations (arbetsgivardeklarationer på individnivå) from Statistics Sweden, covering every employee in Sweden, January 2021 to June 2025, and 4.9 million Platsbanken job advertisements. An employer's exposure is fixed from the 2019 occupations of its workers aged 31–69, scored with the DAIOE index (Engberg et al.), so no young worker's current occupation is needed. Poisson estimates comparing age groups within 104,217 employers, with employer-by-month and age-by-month effects; standard errors clustered by employer.

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